All Things as Wuxing Attractors: MES Universe Curvature Knots and Variable Light Speed

Baoliin (Zaitian) Wu 伍宝林

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1 Physical Proof of Existence

Title: All Things as Wuxing Attractors: MES Universe Curvature Knots and Variable Light Speed
Author: Baoliin (Zaitian) Wu 伍宝林
Date: 2026-10-12
Banknote Serial: `DW58553921`
Paper SHA-256: `fc64d3b3ace4dcd010c7da5cf36ce4d6cc7c09c1888170b25254fb95a5e74665`
Verification URL: https://Zaitian001.github.io/-Self-Preprint-V1.1-/preprints/DW58553921.html

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2 All Things as Wuxing Attractors: MES Universe Curvature Knots and Variable Light Speed

Baoliin (Zaitian) Wu 伍宝林

ORCiD: 0009-0003-2051-5247

Email: zaitian001@gmail.com

Affiliation: People' s Government of Guangdong Province: Guangzhou, CN

Date: October 12, 2026

2.1 Abstract

This paper develops a unified research architecture in which the Modified Einstein Spherical (MES) cosmology architecture describes physical entities as dynamically organized geometric structures within a closed S^3 universe. Its central proposition is that **All Things within the physical domain, including but not limited to all particles, may be investigated as Wuxing Attractors and curvature knots** in the MES Universe Web. The architecture combines a dynamical curvature field, curvature-dependent physical scales, a five-mode state space organized by $\mathbb{Z}_5$, and the Zaitian Quantum Power (ZQP) mechanism.

The paper formulates eight postulates. The first seven address the MES Universe Web, curvature knots, Wuxing Attractor structure, ZQP dynamics, large-scale structure, mass hierarchy, and entanglement. The eighth introduces **ZQP-driven geometric spin emergence** as a research principle: spin is investigated as the rotational transformation property of a stable geometric quantum state, rather than identified a priori with classical internal rotation or with a topological winding number alone.

Within the proposed curvature-scaling architecture, the relations $c(R) = c_0(R/R_0)^{1/2}$, $\hbar(R) = \hbar_0(R_0/R)^{1/2}$, and $m(R) = m_0(R_0/R)^{1/2}$ are examined as linked hypotheses. They preserve $\hbar c$ and mc under the specified scaling, while their interpretation as observable spacetime-dependent constants requires a complete operational and relativistic formulation. A curvature-knot mass formula is retained as a structured hypothesis rather than presented as a completed first-principles derivation.

The paper also develops a geometric interpretation of quantum entanglement, introduces a superquantum postulate as an explicitly conjectural extension, and identifies the mathematical conditions needed to investigate proton spin 1/2 through ZQP-driven geometric dynamics. The model explicitly excludes a sterile neutrino as a physical mode within its proposed five-mode neutrino construction; this is a model choice, not a claim that all possible sterile-neutrino theories have been experimentally ruled out.

Eight postulates, a dimensional-analysis table, and a set of falsifiable research directions are presented. The architecture is a theoretical proposal requiring further mathematical derivation, numerical analysis, and empirical testing; it is not claimed to be experimentally established.

Keywords: MES Universe; All Things; Wuxing Attractors; curvature knots; variable light speed; ZQP; geometric spin emergence; quantum entanglement; neutrino physics; superquantum postulate

2.2 1. Introduction

2.2.1 1.1 The problem of unification

Contemporary physics describes different scales of nature through several highly successful but conceptually distinct architectures. General relativity describes gravitational geometry, the Standard Model describes known elementary particles and their interactions, quantum field theory provides the language of particle dynamics, and cosmology addresses the large-scale evolution of the universe.

These theories are empirically powerful, but several foundational questions remain open. They include the physical origin of particle mass, the relationship between spacetime geometry and quantum properties, the emergence of intrinsic angular momentum, the physical interpretation of entanglement, and the relationship between microscopic and cosmological structures.

The MES cosmology architecture approaches these questions from a geometric starting point. Rather than treating matter as a collection of entities placed on an independently prescribed background, it investigates whether stable matter-like structures can be understood as emergent configurations of a dynamical geometric system.

This is a proposed research program, not a claim that established theories have been displaced.

2.2.2 1.2 The central proposition: All Things as Wuxing Attractors

The central proposition of this paper is:

> **All Things within the physical domain, including but not limited to all particles, may be investigated as Wuxing Attractors embedded in the MES Universe Web.**

Here, *All Things* is deliberately broader than the particle sector. The proposed scope includes, but is not limited to, elementary particles, composite particles, nuclei, atoms, fields, macroscopic matter, stars, black holes, cosmic rays, and large-scale cosmic structures.

This statement is a foundational hypothesis of the MES–Wuxing research program. It does not imply that every listed object has already been shown to correspond to a mathematically established Wuxing Attractor. Such an identification requires an explicit state space, a dynamical law, a stability criterion, and observable consequences for each relevant class of systems.

A Wuxing Attractor (WA) is defined here as a candidate persistent nonlinear state organized by five modes associated with $\mathbb{Z}_5$, with its dynamics governed by a restoring mechanism associated with ZQP.

2.2.3 1.3 The MES–Wuxing–ZQP architecture

The proposed architecture has four interconnected components:

1. **MES Universe Web:** a closed spatial S^3 geometry equipped with a dynamical curvature field.
2. **Curvature knots:** localized or extended geometric configurations proposed as the structural basis of matter-like entities.

3. **Wuxing Attractors:** five-mode nonlinear state structures used to represent persistent and differentiated configurations.
4. **Zaitian Quantum Power:** a proposed universal nonlinear mechanism for restoring, selecting, or stabilizing admissible configurations.

The relationship can be summarized schematically as

$$\text{MES geometry} \longrightarrow \text{ZQP dynamics} \longrightarrow \text{Wuxing Attractors} \longrightarrow \text{observable physical properties.} \quad (1)$$

Equation (1) is an architectural summary, not a derived dynamical equation.

2.2.4 1.4 Logical status of claims

To distinguish foundational assumptions from demonstrated results, this paper uses four categories:

- **Postulate:** a foundational proposition adopted as part of the architecture.
- **Conditional mathematical result:** a conclusion that follows if its stated assumptions hold.
- **Structured hypothesis:** a proposed relationship whose complete derivation or empirical validation remains outstanding.
- **Open problem:** a clearly identified task for future mathematical, numerical, or experimental work.

The distinction is especially important for the mass formula, variable-light-speed interpretation, quantum entanglement, superquantum extension, and proton spin.

2.3 2. Axiomatic Foundation: Overview of the Eight Postulates

The MES–Wuxing architecture is organized around eight postulates. Postulates 1–7 are ontological: they describe the proposed structure of the physical domain. Postulate 8 is methodological: it establishes a research direction for the geometric-dynamical investigation of spin, rather than a fixed claim about the physical world.

2.3.1 2.1 Postulate 1: The MES Universe Web

Statement. The universe is modeled within the MES architecture as a closed spatial S^3 manifold equipped with a dynamical scalar curvature field $R(\mathbf{x}, t)$ and a geometric field $\mathcal{G}_{\mu\nu}(\mathbf{x}, t)$.

The MES Universe Web is defined as

$$\mathcal{W}_{\text{MES}} := \big(S^3, R(\mathbf{x}, t), \mathcal{G}_{\mu\nu}(\mathbf{x}, t)\big)\,. \quad (2)$$

The symbol $\mathcal{W}_{\text{MES}}$ denotes the proposed geometric structure of the model. It is not identical to the observed cosmic web of galaxies, filaments, walls, and voids, although a relationship between them is hypothesized in Postulate 5.

2.3.2 2.2 Postulate 2: All Things as Curvature Structures

Statement. Every physical entity within the proposed scope of MES is investigated as a geometric configuration of the MES Universe Web. Particles are a particular subclass of these entities, and may be represented as curvature knots characterized by their geometric structure and ZQP intervention depth.

$$\mathcal{X}_q \equiv \text{candidate geometric configuration with intervention depth } \mathcal{D}_q. \quad (3)$$

This notation expresses a proposed representation. It does not assert that every physical entity has already been demonstrated to possess a nonzero topological charge or a mathematically knotted field configuration.

2.3.3 2.3 Postulate 3: Wuxing Attractor Structure

Statement. A Wuxing Attractor is represented by a five-mode state space organized by the cyclic group $\mathbb{Z}_5$.

$$\mathcal{H}_{\text{WA}} = \text{span}\,\{|\Phi\rangle, |\Phi\rangle, |\Phi\rangle, |\Phi\rangle, |\Phi\rangle\}. \quad (4)$$

Let $g \in \mathbb{Z}_5$. Define the generative and destructive cyclic operators as

$$S(g) = g + 1 \pmod 5, \qquad K(g) = g + 2 \pmod 5 = S^2(g). \quad (5)$$

The corresponding spectral relation is proposed as

$$\sigma(K) = \sigma(S^2). \quad (6)$$

If σ denotes a spectral mapping that respects operator composition, then $\sigma(S^2) = \{\lambda^2 : \lambda \in \sigma(S)\}$ under the usual finite-dimensional spectral mapping theorem. The shorthand $\sigma(K) = \sigma(S)^2$ is therefore understood in this operator-theoretic sense, not as an unqualified equality between arbitrary sets.

The five modes are abstract dynamical modes. Their assignment to physical species requires independent evidence.

2.3.4 2.4 Postulate 4: ZQP Universality

Statement. The Zaitian Quantum Power (ZQP) is proposed as a universal nonlinear restoring or state-selection mechanism within the MES–Wuxing architecture.

A canonical saturating form is

$$D_q = \lambda_q \tanh\left(\frac{B_q}{s_q}\right). \quad (7)$$

Here B_q is a dimensionless balance variable, $s_q > 0$ is a dimensionless saturation scale, and λ_q has the dimensions assigned to D_q. If D_q is chosen to be dimensionless, then λ_q must also be dimensionless.

The universal feature is the proposed functional form, not the numerical equality of the parameters. Particle- or system-dependent values must be obtained from a specified model or independently measured data.

The use of tanh is a modeling choice within this paper. Its uniqueness as a physical law is not established merely by its oddness, smoothness, and saturation properties.

2.3.5 2.5 Postulate 5: Large-Scale Structure as Curvature Ridges

Statement. The observed cosmic web is hypothesized to be related to the large-scale spatial organization of the MES curvature field.

Let $\Pi_{\text{obs}} : \mathcal{W}_{\text{MES}} \to \mathcal{O}_{\text{cosmic}}$ denote a hypothesized observational projection from the MES field variables to observable cosmic-web quantities. The proposed relationship is

$$\mathcal{W}_{\text{cosmic}} = \Pi_{\text{obs}}[\mathcal{W}_{\text{MES}}] + \varepsilon_{\text{obs}}, \quad (8)$$

where ε_{obs} denotes observational and modeling residuals. The existence and explicit form of Π_{obs} constitute Open Problem 9.

A complete version of this postulate requires a mapping from the MES field variables to observable density fields, gravitational lensing, galaxy distributions, and cosmic expansion observables.

2.3.6 2.6 Postulate 6: Mass Hierarchy from Geometric Structure

Statement. The hierarchy of particle masses is hypothesized to depend on curvature, geometric length scales, and the intervention depth of the relevant Wuxing Attractor.

A representative structured hypothesis is

$$m_q(R_q) = \frac{\hbar_0}{c_0 L_0} \left(\frac{R_0}{R_q} \right)^{1/2} \mathcal{D}_q. \quad (9)$$

The expression is dimensionally a mass when $\mathcal{D}_q$ is dimensionless, L_0 is a length, and $\hbar_0/(c_0 L_0)$ is used as the mass scale.

Equation (9) is not a first-principles derivation of the observed mass spectrum. The quantities L_0, R_0, R_q, and $\mathcal{D}_q$ must be defined and determined independently before the expression can produce parameter-independent predictions.

2.3.7 2.7 Postulate 7: Entanglement as a Geometric Correlation Structure

Statement. Quantum entanglement is investigated through the hypothesis that correlated quantum subsystems may be represented by different components or endpoints of a shared geometric structure in the MES Universe Web.

$$\mathcal{E}_{AB} \nrightarrow \text{shared geometric correlation structure.} \quad (10)$$

The symbol $\nrightarrow$ denotes a proposed interpretation, not an identity proven by the equation.

This postulate does not by itself derive the Born rule, Bell correlations, measurement statistics, or no-signaling. Those results must be demonstrated by an explicit quantum model.

2.3.8 2.8 Postulate 8: ZQP-Driven Geometric Spin Emergence

Statement. Within the MES cosmology architecture, spin is postulated to admit a geometric-dynamical origin: ZQP, acting within the underlying geometric state space, may select and stabilize quantum modes whose transformation properties under spatial rotations give rise to quantized intrinsic angular momentum.

Spin is therefore investigated as a property of stable geometric quantum states, rather than identified a priori with classical internal rotation or with a topological winding number alone.

Let $\mathcal{H}_*$ denote the quantum state space associated with a stable MES–ZQP configuration. The emergence of spin requires a well-defined rotational action on $\mathcal{H}_*$, with generators satisfying the appropriate angular-momentum algebra. The spin quantum number J is then determined by the rotational representation of the resulting quantum state.

The proposed causal structure is

$$\text{MES geometry} \longrightarrow \text{ZQP dynamics} \longrightarrow \text{stable quantum mode} \longrightarrow \text{rotational representation} \longrightarrow J. \quad (11)$$

Scope and status. Postulate 8 establishes a research principle for the geometric-dynamical origin of spin. It does not, by itself, prove that the proton has $J = 1/2$. A complete derivation must construct the relevant state space, define its rotational generators, establish the quantization conditions, and demonstrate why the proton corresponds to the $J = 1/2$ representation.

2.3.9 2.9 Overview of the Eight Postulates

No.	Postulate	Central proposition	Principal proof obligation
1	MES Universe Web	Closed S^3 geometry with a dynamical curvature field	Specify the geometric field equations and their solutions
2	All Things as Curvature Structures	Physical entities are investigated as geometric configurations	Define entity-specific configurations and observables
3	Wuxing Attractor Structure	Five-mode state space organized by $\mathbb{Z}_5$	Establish the dynamical and physical meaning of the modes

No.	Postulate	Central proposition	Principal proof obligation
4	ZQP Universality	Nonlinear restoring or state-selection mechanism	Derive the dynamics and determine the parameters
5	Curvature Ridges	Large-scale structure corresponds to curvature organization	Construct an observational mapping
6	Geometric Mass Hierarchy	Mass depends on curvature and geometric intervention depth	Derive mass scales without fitting known masses
7	Geometric Entanglement	Entanglement may admit a shared geometric representation	Recover quantum statistics and no-signaling
8	Geometric Spin Emergence	ZQP may select stable modes with quantized rotational properties	Derive rotation representations and the proton state

The eight postulates define the foundational scope of the MES–Wuxing architecture. Their mathematical derivations and empirical consequences remain subject to the proof obligations specified throughout this paper.

2.4 3. The MES Universe Web

2.4.1 3.1 Closed spatial geometry

A representative homogeneous metric for a closed spatial geometry is

$$ds^2 = -c(R)^2\,dt^2 + a(t)^2\,d\Omega_3^2. \quad (12)$$

Here, $a(t)$ is the scale factor, $d\Omega_3^2$ is the metric on a unit three-sphere, and $c(R)$ denotes the proposed curvature-dependent propagation speed.

Equation (12) is a model ansatz. If $c(R)$ varies in spacetime, a complete relativistic formulation must specify the metric convention, the role of the lapse function, the operational definition of measured light speed, and the relationship between coordinates and physical clocks.

The expression should not be interpreted as a derivation of variable light speed from the topology of S^3 alone.

2.4.2 3.2 A scalar curvature-field model

For a tractable mathematical investigation, define the dimensionless curvature variable

$$\widetilde{R} = \frac{R}{R_0}. \quad (13)$$

Since the scalar curvature R and reference curvature R_0 both have dimensions of inverse length squared, $\widetilde{R}$ is dimensionless.

A representative effective action is

$$S[\widetilde{R}] = \int d^4x \, \sqrt{-g} \left[\frac{A}{2} g^{\mu\nu} \partial_\mu \widetilde{R} \, \partial_\nu \widetilde{R} - V(\widetilde{R}) \right]. \quad (14)$$

The coefficient A supplies the units required for the action under the chosen unit convention. The potential V must have the corresponding action-density dimensions.

For example, an effective potential may be written as

$$V(\widetilde{R}) = \frac{A\mu^2}{2} \widetilde{R}^2 + \frac{A\eta}{4} \widetilde{R}^4. \quad (15)$$

Here μ has dimensions of inverse length and η has dimensions of inverse length squared in the displayed normalization. Other normalizations are possible, but they must be specified consistently.

Variation with respect to $\widetilde{R}$ yields a scalar-field equation of the form

$$A \, \Box \widetilde{R} + \frac{dV}{d\widetilde{R}} = 0. \quad (16)$$

For the potential in Equation (15), this becomes

$$\Box \widetilde{R} + \mu^2 \widetilde{R} + \eta \widetilde{R}^3 = 0. \quad (17)$$

The sign of the kinetic term and the sign convention for $\Box$ depend on the metric signature and action convention. Equation (17) is a representative scalar-field model, not the complete covariant field equation of the MES cosmology architecture.

The MES architecture may contain additional geometric terms. Those terms must be defined and coupled according to the architecture' s own field equations. The present scalar model should not be confused with a complete derivation of the MES Universe Equation.

2.4.3 3.3 A phenomenological breathing mode

A small oscillatory perturbation of the scale factor may be represented as

$$a(t) = a_0 \left[1 + \epsilon \cos(\omega_b t) \right], \qquad |\epsilon| \ll 1. \quad (18)$$

This expression describes a phenomenological breathing mode. The associated scalar curvature cannot, in general, be specified by a universal proportionality to the scale factor without choosing the spacetime metric, curvature convention, and field equations.

Accordingly, the curvature oscillation must be calculated from the selected MES solution rather than assumed to follow from Equation (18) alone.

2.5 4. Curvature-Dependent Physical Scales and Variable Light Speed

2.5.1 4.1 Proposed curvature scaling

The MES scaling hypothesis adopted in this version is

$$c(R) = c_0 \left(\frac{R}{R_0} \right)^{1/2}. \quad (19)$$

The associated reduced Planck constant is proposed to scale as

$$\hbar(R) = \hbar_0 \left(\frac{R_0}{R} \right)^{1/2}. \quad (20)$$

The mass scale is proposed to obey

$$m(R) = m_0 \left(\frac{R_0}{R} \right)^{1/2}. \quad (21)$$

These relations require $R/R_0 > 0$ in the displayed real-valued form. If the intended curvature domain includes zero or negative scalar curvature, an extension of the scaling law must be specified.

Equations (19)–(21) are linked hypotheses within the MES architecture. They are not derived from the existence of a closed S^3 manifold alone.

2.5.2 4.2 Invariant products under the proposed scaling

Multiplying Equations (19) and (20) gives

$$\hbar(R)\, c(R) = \hbar_0\, c_0. \quad (22)$$

Likewise, Equations (19) and (21) give

$$m(R)\, c(R) = m_0\, c_0. \quad (23)$$

These two identities follow algebraically from the specified scaling relations. They do not establish that either $\hbar$ or c varies in a physically measurable way.

A physical interpretation requires a complete theory of clocks, rulers, wave propagation, local measurements, and dimensionless observables. In particular, a variation of dimensional constants

cannot be established solely by assigning position-dependent values to their numerical representations.

2.5.3 4.3 Propagation time

For a prescribed path γ with spatial line element ds, the corresponding coordinate propagation time in the proposed model may be expressed as

$$t_{AB} = \int_{\gamma: A \to B} \frac{ds}{c(R(s))}. \quad (24)$$

Equation (24) is a propagation-time expression under the assumed local-speed rule. It does not independently establish that a signal can exceed the locally measured light speed, violate a causal cone, or transmit information superluminally.

Those conclusions require the complete metric and causal structure of the MES model.

2.5.4 4.4 Propagation regimes

The proposed law implies the following algebraic regimes:

Curvature ratio	Predicted ratio c/c_0	Interpretation within the ansatz
$R/R_0 > 1$	Greater than 1	Higher coordinate propagation speed
$R/R_0 = 1$	Equal to 1	Reference regime
$0 < R/R_0 < 1$	Between 0 and 1	Lower coordinate propagation speed
$R/R_0 \to 0^+$	Approaches 0	Singular or limiting regime requiring further analysis

Numerical assignments of extreme curvature ratios to black-hole interiors, active galactic nuclei, interstellar space, or cosmic voids are not included as established physical values. They require a specified curvature definition, observational constraints, and a consistent relativistic interpretation.

2.5.5 4.5 Observational requirements

The variable-light-speed hypothesis must be tested through quantities that can be measured independently of arbitrary unit conventions. Relevant investigations include:

- time-of-flight comparisons between signals from a common source;
- gravitational lensing and propagation delays;
- consistency of electromagnetic and gravitational-wave observations;
- dimensionless atomic and particle observables across environments;
- constraints from local Lorentz invariance and established tests of relativity.

The theory must predict quantitative departures from the relevant baseline models before observational agreement or disagreement can be assessed.

2.6 5. Wuxing Attractors as Geometric Structures

2.6.1 5.1 State representation

A normalized state in the proposed five-mode space may be written as

$$|\Psi_{\mathrm{WA}}\rangle = \sum_{q\in\{\ ,\ ,\ ,\ ,\ \}} \alpha_q |\Phi_q\rangle, \qquad \sum_q |\alpha_q|^2 = 1. \quad (25)$$

The normalization is the standard Hilbert-space condition under the assumption that the five basis states are orthonormal.

The five labels represent abstract modes in this model. Their interpretation as specific particle species or physical interactions is not implied by the notation alone.

2.6.2 5.2 Balance functional

A candidate balance functional is

$$B_{\mathrm{WA}} = \sum_q w_q |\alpha_q|^2. \quad (26)$$

For B_{WA} to be dimensionless, the weights w_q must be dimensionless. If the functional is intended to represent a physical observable, its relation to measurable quantities must be specified.

The condition $B_{\mathrm{WA}} = 0$ defines a balance surface in the state space. It does not automatically establish a dynamical attractor.

2.6.3 5.3 Nonlinear Symmetry Anchor Zero

Definition 1. A Nonlinear Symmetry Anchor Zero (NSAZ) is a designated reference state or balance surface at which the selected restoring variable vanishes.

Let $\psi(B)$ be a candidate restoring function satisfying

$$\psi(-B) = -\psi(B), \qquad \psi(0) = 0, \qquad \sup_B |\psi(B)| < \infty. \quad (27)$$

A canonical choice is

$$\psi(B) = \tanh\left(\frac{B}{s_{\mathrm{sat}}}\right), \qquad s_{\mathrm{sat}} > 0. \quad (28)$$

Equation (28) is smooth, odd, bounded, and monotonic. These properties make it a useful candidate for nonlinear feedback. They do not prove that tanh is the unique function permitted by MES or by ZQP.

2.6.4 5.4 Attractor conditions

A state X_* is a dynamical attractor only if a specified evolution law has the required stability properties near X_*. A general candidate evolution equation is

$$\frac{dX}{d\tau} = F_{\text{geom}}[X] + F_{\text{ZQP}}[X] + F_{\mathbb{Z}_5}[X]. \quad (29)$$

Equation (29) is an abstract decomposition. Each term requires a definition on the state space, including its domain, units, and regularity conditions.

To demonstrate attraction, one must identify a stationary solution and analyze the linearized dynamics or an appropriate nonlinear stability criterion. A balance condition alone is insufficient.

2.7 6. Mass as Curvature-Driven Emergence

2.7.1 6.1 Structured mass hypothesis

The proposed mass formula is

$$m_q(R_q) = \frac{\hbar_0}{c_0 L_0} \left(\frac{R_0}{R_q}\right)^{1/2} \mathcal{D}_q. \quad (30)$$

The prefactor $\hbar_0/(c_0 L_0)$ has dimensions of mass.

The dimensions are consistent if:

- $\hbar_0$ has dimensions of action;
- c_0 has dimensions of length per time;
- L_0 has dimensions of length;
- R_0/R_q is dimensionless;
- $\mathcal{D}_q$ is dimensionless.

Then $\hbar_0/(c_0 L_0)$ has dimensions of mass.

Equation (30) is a **structured hypothesis**. It expresses a proposed relationship between mass, curvature, a characteristic length, and ZQP intervention depth. It is not a first-principles derivation of the observed particle spectrum.

2.7.2 6.2 Required derivations

Four open problems remain central to the mass formula.

Curvature scales (Open Problem 2). Determine L_0 and R_0 from the MES field equations or independent observations rather than fitting known particle masses.

Intervention depth (Open Problem 3). Derive $\mathcal{D}_q$ from the specified ZQP dynamics.

Intervention-depth assignment (Open Problem 3). Explain why different physical entities occupy different intervention-depth regimes.

Mass spectrum (Open Problem 4). Produce parameter-independent mass predictions with uncertainty estimates and explicit comparison against measured values.

2.7.3 6.3 Illustrative mass hierarchy

Candidate entity	Proposed role of $\mathcal{D}_q$	Status
Neutrinos	Small intervention depth	Requires independent derivation
Electron	Intermediate intervention depth	Requires independent derivation
Proton	Larger effective intervention depth	Requires independent derivation
Black holes	Macroscopic geometric configuration	Requires a separate field and stability treatment

Numerical values for the intervention depths must not be selected merely to reproduce known masses and then treated as independent predictions. Such a procedure would be parameter fitting, not a first-principles derivation.

2.7.4 6.4 Relation to the MES scaling

If the mass scaling in Equation (21) is adopted, the mass ratio between two positive-curvature regimes is

$$\frac{m(R_2)}{m(R_1)} = \left(\frac{R_1}{R_2}\right)^{1/2}. \quad (31)$$

This is an algebraic consequence of the assumed scaling law. Its physical validity depends on whether the model defines a meaningful operational comparison of masses across the relevant regimes.

2.8 7. ZQP as a Universal Restoring Mechanism

2.8.1 7.1 Coupling function

The ZQP coupling is represented by the common functional form

$$D_q = \lambda_q \tanh\left(\frac{B_q}{s_q}\right). \quad (32)$$

Here, B_q and s_q are dimensionless. The dimensions of D_q and λ_q must match.

The proposed universality is functional: the same nonlinear form is used across different systems. The parameters may differ and must be specified for each application.

2.8.2 7.2 Candidate parameter hierarchy

System	λ_q	s_q	Interpretation
Neutrino sector	To be determined	To be determined	Candidate weakly coupled mode
Electron	To be determined	To be determined	Candidate charged-lepton mode
Proton	To be determined	To be determined	Candidate composite-hadron mode
Black hole	To be determined	To be determined	Requires a macroscopic field description

The table intentionally leaves numerical values unspecified. Arbitrary orders of magnitude would not be physically meaningful without a derivation or a clearly identified fit.

2.8.3 7.3 Candidate restoring-rate law

A proposed curvature-dependent rate is

$$\Gamma_q(R) = \gamma_0 \lambda_q \left(\frac{R}{R_0} \right)^{3/2}. \quad (33)$$

For Γ_q to have dimensions of inverse time, $\gamma_0 \lambda_q$ must have those dimensions if the curvature ratio is dimensionless.

A representative stochastic equation is

$$\frac{dB_q}{dt} = -\Gamma_q(R)\, \psi(B_q) + \xi_q(t). \quad (34)$$

Here, $\xi_q(t)$ represents a noise term whose statistical properties must be specified.

For small B_q, the approximation $\tanh(B_q/s_q) \approx B_q/s_q$ gives

$$\frac{dB_q}{dt} \approx -\frac{\Gamma_q(R)}{s_q} B_q + \xi_q(t). \quad (35)$$

The deterministic linearized equation has a restoring tendency when $\Gamma_q(R)/s_q > 0$. The scaling $\Gamma_q \propto R^{3/2}$, however, remains a structured hypothesis until derived from a specified ZQP field equation or independently tested.

2.8.4 7.4 Open quantum dynamics

The broader MES architecture considers curvature-coupled open quantum dynamics. A standard candidate representation for a Markovian reduced density operator is the Lindblad equation

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2}\{L_j^\dagger L_j, \rho\} \right). \quad (36)$$

Equation (36) is a standard open-quantum-system form. To use it as an MES prediction, the Hamiltonian H, the operators L_j, and their dependence on the MES curvature field must be derived or explicitly modeled.

A derivation from microscopic dynamics requires assumptions concerning the system–environment split, coupling strength, bath correlations, and relevant time scales. Curvature dependence alone does not guarantee a Lindblad equation.

2.9 8. Neutrino Wuxing Attractors

Within the proposed MES–Wuxing architecture, the extreme smallness of neutrino masses is tentatively recast not as an isolated free parameter but as the ZQP-driven stabilization of neutrino curvature knots at a shallow intervention depth $\mathcal{D}_q$ within the MES Universe Web. Correspondingly, flavor oscillations are reinterpretable as ZQP-driven transitions among the five Wuxing modes ($\mathbb{Z}_5$). This is offered as a structured hypothesis whose falsifiable consequences remain to be derived.

2.9.1 8.1 Active neutrino states

For the purpose of exploring a five-mode Wuxing representation, three modes may be associated with the active neutrino flavors:

$$|\nu\rangle \equiv |\nu_e\rangle, \qquad |\nu\rangle \equiv |\nu_\mu\rangle, \qquad |\nu\rangle \equiv |\nu_\tau\rangle. \quad (37)$$

The two remaining labels require independent physical definitions. They cannot be identified with experimentally established particle species solely because the mathematical state space has five modes.

2.9.2 8.2 Exclusion of a sterile neutrino mode

The present MES–Wuxing model explicitly excludes the sterile-neutrino hypothesis as a physical mode in its proposed neutrino state space; this is a model choice and not a claim that every possible sterile-neutrino model has been experimentally excluded.

The two remaining modes $|\nu\rangle$ and $|\nu\rangle$ are presently defined as **geometric phase modes** –internal degrees of freedom associated with the curvature field and ZQP coupling, rather than independent particle species. Their possible physical interpretation beyond the present model remains an open problem.

In particular, the fifth-mode construction must not be used to reintroduce a sterile neutrino under another name without specifying the relevant observable distinctions and experimental consequences.

2.9.3 8.3 Mass and coherence length

A candidate coherence-length definition is

$$L_q(R) = \frac{\hbar(R)\, c(R)}{\Lambda_q\, |B_\nu^{(q)}|}. \quad (38)$$

Here, Λ_q is an energy scale and $B_\nu^{(q)}$ is a dimensionless balance parameter. Since $\hbar c$ has dimensions of energy times length, Equation (38) has dimensions of length.

If one defines the associated energy scale by

$$E_q = \frac{\hbar(R)\, c(R)}{L_q(R)}, \quad (39)$$

then substituting Equation (38) yields $E_q = \Lambda_q |B_\nu^{(q)}|$. This is an algebraic consequence of the definition, not an independent derivation of a neutrino mass.

To identify E_q with $m_{\nu_q} c^2$, the relevant energy–mass convention and the dynamical origin of the coherence length must be established.

2.9.4 8.4 Neutrino mass constraints

Neutrino masses are constrained by oscillation measurements, cosmological observations, and direct kinematic searches. A proposed MES spectrum must be compared with the relevant mass-squared differences and bounds, while distinguishing mass eigenstates from flavor eigenstates.

Any numerical spectrum adopted in a future version should be accompanied by current experimental sources, uncertainties, and a clear statement of which parameters are fitted and which are predicted.

2.9.5 8.5 No claimed resolution of sterile-neutrino anomalies

Because the sterile-neutrino hypothesis is excluded from the physical mode content of the present model, this paper does not claim that MicroBooNE, STEREO, or other experiments have excluded all sterile-neutrino models, nor does it claim that their results have been explained by a verified curvature-suppression mechanism.

Any future attempt to account for neutrino anomalies must formulate a separate quantitative model, specify its predictions, and test them against existing experimental likelihoods and constraints.

2.9.6 8.6 Mass ordering

The conditional relation

$$\text{normal mass ordering} \iff R_{\mathrm{QCD}} > R_0 \quad (40)$$

is not retained as a theorem. A mass-ordering conclusion does not follow from an inequality between curvature scales unless a defined mass-eigenvalue equation establishes the implication in both directions.

A valid derivation must connect the curvature field to the neutrino mass matrix, derive the eigenvalues, and demonstrate how the ordering follows without assuming the desired outcome.

2.10 9. Nucleon Geometry and the Emergence of Spin

2.10.1 9.1 Nucleon state representation

A nucleon candidate state in the five-mode architecture may be written as

$$|N\rangle = \sum_q c_q |\Phi_q\rangle, \qquad \sum_q |c_q|^2 = 1. \quad (41)$$

A possible assignment of abstract modes to quark- and field-related degrees of freedom is a modeling choice, not a derivation of QCD. Any such assignment must preserve the relevant quantum numbers and avoid treating the five Wuxing labels as a replacement for the established color, flavor, and spin structure of the Standard Model.

2.10.2 9.2 Why topology alone does not prove proton spin

For an appropriately defined SU(3)-valued effective field $U : S^3 \to SU(3)$, the homotopy relation

$$\pi_3(SU(3)) \cong \mathbb{Z} \quad (42)$$

permits integer-valued topological sectors.

With a suitable orientation and normalization, one may define a winding number

$$B[U] = \frac{1}{24\pi^2} \int_{S^3} \mathrm{Tr}\left[(U^{-1}dU)^3\right] \in \mathbb{Z}. \quad (43)$$

The exact overall sign depends on conventions for the trace and orientation.

Equations (42) and (43) establish the mathematical possibility of classifying suitable field configurations by an integer topological charge. They do not establish that the proton spin equals one-half, and they do not establish that every SU(3) color-gauge configuration carries the physical baryon number.

The physical interpretation of $B[U]$ depends on the field theory in which U is defined. In a chiral effective theory, topological winding can be related to baryon number under specific conditions. This relationship cannot be transferred automatically to an arbitrary MES geometric field.

2.10.3 9.3 ZQP-driven geometric spin emergence

Postulate 8 reverses the question. Rather than presupposing a proton spin of one-half and identifying it with a winding number, it asks whether ZQP-driven geometric dynamics can produce stable quantum states with the appropriate rotational transformation properties.

Let $\mathcal{C}_B$ denote a configuration space restricted to a well-defined topological sector, and let $\gamma_{2\pi}$ represent a path corresponding to a full spatial rotation of a candidate configuration. A complete

quantization must determine whether this path is contractible and what phase the corresponding quantum state acquires.

If the model independently establishes

$$\Psi[\gamma_{2\pi}(U)] = -\Psi[U], \quad (44)$$

then the state transforms with a minus sign under a 2π rotation. In an ordinary quantum theory with the appropriate rotation representation, this is consistent with half-integer spin.

However, Equation (44) is a condition to be derived from the model' s configuration-space topology and quantization rules. It does not follow from the integer winding number alone.

2.10.4 9.4 The distinction between half-integer spin and spin one-half

The angular momentum algebra permits

$$J \in \{\tfrac{1}{2}, \tfrac{3}{2}, \tfrac{5}{2}, ...\} \quad (45)$$

for half-integer-spin representations.

Consequently, even a successful derivation of the minus sign in Equation (44) would not by itself select $J = 1/2$. To establish the proton' s specific spin, the theory must additionally derive the relevant energy spectrum, quantum numbers, and state-selection mechanism.

2.10.5 9.5 Proof obligations for the proton

A complete MES-ZQP derivation requires the following steps.

Step	Content	Status
1	Field definition	Open
2	Topological classification	Partially known ($\pi_3(SU(3)) \cong \mathbb{Z}$)
3	Rotational action	Open
4	Quantization	Open
5	State selection ($J = 1/2$)	Open
6	Empirical consistency	Open

The geometric-spin proposal is therefore a defined research program, not a completed proof.

2.10.6 9.6 The proton-neutron mass difference

The proton–neutron mass difference is an established observable, but it is not derived by specifying a curvature knot alone. A MES explanation would need to produce a quantitative difference between the corresponding states, account for electromagnetic contributions and quark-mass effects, and reproduce the measured sign and magnitude within uncertainty.

This remains an open problem.

2.11 10. Quantum Entanglement and the Geometric Correlation Hypothesis

2.11.1 10.1 A representative entangled state

A two-subsystem singlet state may be represented as

$$|\Psi_{\rm ent}\rangle = \frac{1}{\sqrt{2}}\left(|\nu_-\rangle_A|\nu_+\rangle_B - |\nu_+\rangle_A|\nu_-\rangle_B\right). \quad (46)$$

This is a standard quantum-state form under the assumption that the two basis states represent the appropriate orthogonal degrees of freedom. The expression does not establish that the subsystems are physically connected by a single geometric knot.

2.11.2 10.2 Geometric hypothesis

The proposed MES interpretation is

$$\text{entanglement} \rightsquigarrow \text{shared geometric correlation structure.} \quad (47)$$

The hypothesis is that the correlated subsystems may be represented as parts or endpoints of a common geometric structure. To turn this into a physical model, the relationship between the geometric state and the quantum density operator must be explicitly defined.

In particular, the model must reproduce quantum measurement probabilities and the correlations observed in Bell-type experiments.

2.11.3 10.3 A candidate geometric kernel

A possible positive elliptic operator on S^3 can be written as

$$\hat{A} = -\Delta_{S^3} + \mu_K^2, \qquad \mu_K > 0. \quad (48)$$

Here, $-\Delta_{S^3}$ is the nonnegative Laplace–Beltrami operator on the specified spatial geometry, and μ_K has dimensions of inverse length.

The corresponding Green operator is

$$\hat{K} = \hat{A}^{-1}. \quad (49)$$

Let the eigenfunctions satisfy

$$-\Delta_{S^3}\phi_k = \kappa_k^2\phi_k. \quad (50)$$

For an orthonormal basis under the selected volume measure,

$$\int_{S^3} \phi_k(\mathbf{x})\,\phi_{k'}^*(\mathbf{x})\,dV = \delta_{kk'}. \quad (51)$$

The formal kernel is then

$$K(\mathbf{x}, \mathbf{x}') = \sum_k \frac{\phi_k(\mathbf{x})\, \phi_k^*(\mathbf{x}')}{\kappa_k^2 + \mu_K^2}. \quad (52)$$

The dimensions of the kernel depend on the normalization of the eigenfunctions and the measure. The kernel also depends on the chosen metric and boundary conditions.

Equations (48)–(52) define a possible mathematical tool for representing spatial correlations. They do not demonstrate that quantum entanglement is mediated by this kernel, nor that the kernel determines a quantum measurement outcome.

2.11.4 10.4 Nonlocal correlation and causality

The idea of "instantaneous activation" is retained only as a geometric hypothesis about nonlocal correlation. It is not asserted to be a physical signal traveling instantaneously between spatially separated systems.

A valid MES account must demonstrate that its predicted measurement statistics reproduce the relevant quantum correlations and satisfy the no-signaling condition. It must also specify how the model' s causal structure is related to operationally measured propagation speeds.

Open Problem 7. Derive the measurement statistics of the WA architecture, prove the no-signaling condition under clearly specified assumptions, and establish consistency with the causal structure of the MES model.

2.11.5 10.5 Relation to Bell inequalities

For the standard CHSH experiment, the local-hidden-variable bound is 2, while quantum theory predicts a maximum value of $2\sqrt{2}$ for the usual quantum-mechanical correlations under the standard assumptions.

A proposed MES extension must not conflate a mathematical parameter with an experimentally measured CHSH correlation. Any claimed departure from the standard quantum bound requires a complete specification of the observables, probabilities, normalization, and experimental protocol.

2.12 11. The Superquantum Extension as a Separate Conjecture

The superquantum expression is not counted as an additional foundational postulate in this version. The eight-postulate architecture is preserved, while the proposed extension is formulated separately as a conjectural research hypothesis.

2.12.1 11.1 The superquantum conjecture

Superquantum Conjecture (conjectural extension). The MES–ZQP architecture investigates whether curvature-dependent geometric dynamics can define an effective correlation structure that differs from standard quantum predictions under a precisely specified extension of the measurement model.

The candidate expression is

$$S_{\mathrm{ZQP}} = 2\sqrt{2}\left(1 + \frac{\lambda_q R}{s_q R_0}\right). \quad (53)$$

Status of the superquantum expression. The expression $S_{\mathrm{ZQP}} = 2\sqrt{2}(1 + \lambda_q R/(s_q R_0))$ is presented only as a **mathematical placeholder** for a possible curvature-dependent correlation. It is not claimed that this expression is dimensionally identical to a CHSH parameter, nor that it violates the Tsirelson bound in any operational sense. A consistent derivation from the MES measurement model remains an open problem.

For the ratio in parentheses to be dimensionless, λ_q and s_q must have compatible dimensions, and R/R_0 is dimensionless. If λ_q and s_q are both dimensionless, the expression is dimensionally consistent.

2.12.2 11.2 Mathematical and empirical requirements

If Equation (53) is intended to represent an actual CHSH correlation parameter, it must be reconciled with the definitions of the measured observables and the probability structure of the theory.

In standard quantum theory, a CHSH parameter formed from the usual dichotomic observables is bounded by $2\sqrt{2}$. Therefore, if the correction factor in Equation (53) is positive and nonzero, the expression exceeds the standard quantum bound. Such a result cannot be presented as an ordinary quantum-mechanical prediction without further qualification.

A superquantum interpretation would require, at minimum:

1. a precise definition of the proposed observables;
2. a normalized probability model;
3. explicit predictions for measurement outcomes;
4. a demonstration of no-signaling, or a clearly stated alternative;
5. a causal and relativistic interpretation;
6. a testable distinction from standard quantum theory.

Open Problem 6. Derive or revise Equation (53) from a fully specified MES-ZQP measurement model, and determine whether the resulting theory is mathematically consistent, operationally meaningful, and empirically distinguishable from standard quantum mechanics.

2.13 12. Cosmic Rays as Curvature-Associated Excitations

2.13.1 12.1 State representation

A candidate state representation for a cosmic-ray sector is

$$|\Phi_{\mathrm{CR}}(t)\rangle = \sum_q \beta_q(t)|\Phi_q\rangle. \quad (54)$$

The coefficients $\beta_q(t)$ must be determined by a specified evolution equation and appropriate normalization.

2.13.2 12.2 A candidate curvature-dependent energy scale

A dimensionally consistent candidate energy scale is

$$E_{\mathrm{CR}}(R) = \frac{\hbar_0 c_0}{L_0} \left(\frac{R}{R_0} \right)^{1/2} \mathcal{E}_{\mathrm{CR}}. \quad (55)$$

Here, L_0 is a length and $\mathcal{E}_{\mathrm{CR}}$ is dimensionless. Equation (55) is a phenomenological form; it does not by itself derive the cosmic-ray energy spectrum or its acceleration mechanism.

The coefficient and curvature dependence must be compared against measured spectra, source distributions, and propagation effects.

2.13.3 12.3 Neutrino production and flavor ratios

High-energy neutrino flavor ratios depend on production mechanisms, propagation, oscillations, and the detector response. A schematic Wuxing-mode representation can be written as

$$\mathbf{f}_\nu = \mathcal{F}_{\mathrm{WA}}(\epsilon, \theta, \mathcal{P}), \quad (56)$$

where ϵ denotes model parameters, θ denotes mixing parameters, and $\mathcal{P}$ denotes source and propagation assumptions.

The function $\mathcal{F}_{\mathrm{WA}}$ must be specified before the model can predict flavor ratios. No numerical ratio is claimed by Equation (56) alone.

2.14 13. Falsifiable Predictions and Research Tests

The MES–Wuxing architecture should be evaluated through predictions that can fail, rather than through qualitative correspondence alone.

ID	Research prediction or hypothesis	Required test	Current status
P1	Curvature-dependent propagation law	Quantitative time-of-flight, lensing, and local-invariance tests	Hypothesis; full operational model required
P2	Curvature-dependent mass scaling	Independent comparison across defined physical regimes	Structured hypothesis
P3	ZQP restoring-rate scaling	Derivation and controlled simulation or experiment	Open

ID	Research prediction or hypothesis	Required test	Current status
P4	Five-mode Wuxing state structure	Independent identification of modes and observable consequences	Open
P5	ZQP-driven geometric spin emergence	Derive rotational generators, quantization, and state selection	Open
P6	Geometric representation of entanglement	Recover quantum statistics and no-signaling	Open
P7	Superquantum extension	Fully specified correlation experiment and consistency tests	Conjectural
P8	MES mapping to cosmic structure	Compare predicted curvature observables with astronomical data	Open

2.14.1 13.1 Requirements for a meaningful test

A testable prediction must specify:

- the mathematical model and parameter values;
- the baseline theory against which it is compared;
- the measured observable;
- the expected uncertainty;
- the conditions under which the prediction would be rejected.

A statement that a phenomenon "may be explained geometrically" is not yet a falsifiable prediction. The model must calculate a measurable quantity.

2.14.2 13.2 Numerical validation

Numerical simulations may be useful for exploring the stability of Wuxing Attractors, ZQP feedback, and curvature-associated modes. Such simulations must distinguish between:

- results built into the equations by construction;
- results emerging from the dynamics without being imposed;
- parameter fitting to known observations;
- genuinely out-of-sample predictions.

For example, a numerical model designed to produce five stable modes does not independently establish that nature has exactly five corresponding physical modes. That correspondence requires

a separate empirical argument.

2.15 14. Discussion

2.15.1 14.1 What the architecture proposes

The MES–Wuxing architecture proposes that:

- the MES Universe Web provides a geometric foundation for investigating physical structures;
- physical entities, including but not limited to particles, may be represented as geometric configurations;
- Wuxing Attractors provide a five-mode nonlinear representation;
- ZQP may act as a common restoring or state-selection mechanism;
- mass hierarchy may depend on curvature and geometric intervention depth;
- quantum entanglement may admit a geometric representation;
- spin may emerge from stable geometric quantum states selected by ZQP dynamics;
- curvature-dependent physical scales may offer a route toward a unified geometric description.

These are propositions of the research program, not a list of experimentally established results.

2.15.2 14.2 What the architecture does not claim

This paper does not claim that:

- the MES architecture has experimentally replaced general relativity or the Standard Model;
- all physical entities have already been proven to be Wuxing Attractors;
- the mass formula has been derived from first principles;
- the curvature-dependent light-speed relation has been experimentally verified;
- topology alone proves proton spin 1/2;
- the geometric entanglement hypothesis has been proven to reproduce all quantum measurement statistics;
- the superquantum expression has been experimentally verified;
- the sterile-neutrino hypothesis has been excluded from all possible physical theories.

The sterile-neutrino model choice in Section 8 applies to the proposed MES–Wuxing mode construction only.

2.15.3 14.3 The significance of the eighth postulate

The eighth postulate changes the direction of the spin problem. Instead of assuming that spin is a pre-existing quantum number and searching for a topological label, the MES–ZQP program asks whether geometric dynamics can generate stable states whose rotational transformation properties correspond to spin.

This is a more foundational question, but it also imposes a more demanding proof requirement. The model must derive the quantum state space, the rotational action, the angular-momentum algebra, and the particular representation associated with the proton.

ZQP can be proposed as a mechanism for selecting and stabilizing geometric modes. It cannot be identified with the origin of spin in a demonstrated sense until the required dynamics and quantization have been supplied.

2.15.4 14.4 The distinction between architecture and derivation

The MES architecture is the theoretical starting point of this paper. The purpose of the mathematical sections is not to presuppose that every relation has already been derived, but to identify the equations, definitions, and tests needed to make the architecture precise.

A complete future treatment should integrate the curvature field equations, ZQP dynamics, state-space structure, and observable predictions into a consistent mathematical model. It should also establish the assumptions under which the model reduces to established physical theories in their tested domains.

2.16 15. Open Problems

The following is the consolidated list of open problems referenced throughout the paper.

Open Problem 1: Curvature dynamics. Specify the complete MES field equations, including the definitions and couplings of the relevant geometric variables.

Open Problem 2: Curvature scales. Derive R_0, L_0, and other reference scales from the model or independent observations.

Open Problem 3: Intervention depth. Derive $\mathcal{D}_q$ from ZQP dynamics and establish its relation to measurable particle properties.

Open Problem 4: Mass spectrum. Obtain parameter-independent predictions for masses and mass ratios.

Open Problem 5: ZQP rate. Derive the proposed $\Gamma_q \propto R^{3/2}$ scaling from a specified dynamical equation.

Open Problem 6: Superquantum extension. Establish whether the conjectural correlation expression has a consistent operational interpretation and whether it can be tested without contradicting the explicitly stated assumptions of the model.

Open Problem 7: Entanglement. Derive the measurement statistics, no-signaling condition, and causal interpretation of the proposed geometric correlation structure.

Open Problem 8: Proton spin. Construct the MES–ZQP state space, derive the rotational representation, and demonstrate why the proton corresponds to $J = 1/2$.

Open Problem 9: Cosmic structure. Develop an observational map between the MES curvature field and large-scale cosmic observables.

Open Problem 10: Empirical discrimination. Identify quantitative predictions that distinguish MES from established theories and competing extensions.

2.17 16. Conclusion

This paper presents the MES–Wuxing research architecture, centered on the proposition that **All Things within the physical domain, including but not limited to all particles, may be**

investigated as Wuxing Attractors and curvature-associated structures in the MES Universe Web.

The architecture now contains eight postulates. The new eighth postulate, ZQP-Driven Geometric Spin Emergence, establishes a research principle for investigating spin as a rotational property of stable geometric quantum states. It reframes the proton-spin question as a problem of geometric dynamics, quantization, and state selection.

The curvature-scaling relations are expressed as

$$c(R) = c_0 \left(\frac{R}{R_0}\right)^{1/2}, \qquad \hbar(R) = \hbar_0 \left(\frac{R_0}{R}\right)^{1/2}, \qquad m(R) = m_0 \left(\frac{R_0}{R}\right)^{1/2}. \quad (57)$$

Their algebraic consequences include

$$\hbar(R)\, c(R) = \hbar_0\, c_0, \qquad m(R)\, c(R) = m_0\, c_0. \quad (58)$$

These identities follow from the proposed scaling relations. Their physical interpretation remains contingent on a complete operational formulation.

The mass formula remains a structured hypothesis. The geometric interpretation of entanglement remains a hypothesis requiring recovery of quantum statistics and no-signaling. The superquantum expression remains a conjectural extension. The proton-spin explanation remains an open proof program rather than an established theorem.

The central contribution of this paper is therefore the consolidation of a broad geometric research architecture with clearer proof obligations, corrected dimensional relations, and a new principle for the geometric-dynamical investigation of spin.

$$\boxed{\text{The universe is investigated as the MES Universe Web.}} \quad (59)$$

$$\boxed{\text{All Things may be investigated as Wuxing Attractors.}} \quad (60)$$

$$\boxed{\text{ZQP-driven geometric spin emergence is a foundational research hypothesis.}} \quad (61)$$

2.18 Appendix A. Symbol and Dimensional Table

Symbol	Meaning	Dimension
R	Scalar curvature	$[L]^{-2}$
R_0	Reference curvature	$[L]^{-2}$
$\widetilde{R}$	Dimensionless curvature	1
$c(R)$	Proposed curvature-dependent speed	$[L][T]^{-1}$
c_0	Reference speed	$[L][T]^{-1}$

Symbol	Meaning	Dimension
$\hbar(R)$	Proposed reduced Planck constant	$[M][L]^2[T]^{-1}$
$\hbar_0$	Reference reduced Planck constant	$[M][L]^2[T]^{-1}$
m_q	Particle mass	$[M]$
m_0	Reference mass	$[M]$
L_0	Characteristic geometric length	$[L]$
L_q	Coherence length	$[L]$
$\mathcal{D}_q$	ZQP intervention depth	1, as defined here
B_q	Balance variable	1
s_q	Saturation scale	1
λ_q	ZQP coupling amplitude	Same as D_q
Γ_q	Restoration rate	$[T]^{-1}$
γ_0	Rate normalization in Equation (33)	Depends on the dimensions of λ_q
ω_b	Breathing-mode frequency	$[T]^{-1}$
ϵ	Dimensionless perturbation amplitude	1
μ	Inverse-length parameter in the effective potential	$[L]^{-1}$
η	Cubic-potential coefficient in Equation (15)	$[L]^{-2}$
μ_K	Kernel inverse-length scale	$[L]^{-1}$
κ_k	Spatial eigenvalue wave number	$[L]^{-1}$
$B[U]$	Topological winding number	1
J	Spin quantum number	Dimensionless
$\hat{\mathbf{J}}$	Angular-momentum operator	$[M][L]^2[T]^{-1}$
Λ_q	Energy scale	$[M][L]^2[T]^{-2}$

2.18.1 A.1 Dimensional checks

The proposed mass scale in Equation (30) has the required mass dimension:

$$\left[\frac{\hbar_0}{c_0 L_0}\right] = [M]. \quad (62)$$

The curvature ratio is dimensionless:

$$\left[\frac{R}{R_0}\right] = 1. \quad (63)$$

The coherence length in Equation (38) has the required length dimension:

$$\left[\frac{\hbar c}{\Lambda_q}\right] = [L]. \quad (64)$$

For the kernel in Equation (49), both terms in the operator must have the same dimension:

$$[-\Delta_{S^3}] = [\mu_K^2] = [L]^{-2}. \quad (65)$$

The superquantum expression in Equation (53) is dimensionless only if the ratio λ_q/s_q is dimensionless:

$$\left[\frac{\lambda_q R}{s_q R_0}\right] = 1. \quad (66)$$

These checks establish dimensional compatibility under the stated definitions. They do not establish the physical validity of the proposed relations.

2.19 Appendix B. Claim Classification

Claim	Classification	What remains necessary
Closed S^3 geometry as the MES starting point	Postulate	Specify its full field dynamics
All Things as Wuxing Attractors	Foundational hypothesis	Demonstrate entity-specific mappings
$\mathbb{Z}_5$ five-mode representation	Mathematical construction	Establish physical mode assignments
Universal ZQP restoring mechanism	Postulate and model proposal	Derive dynamics and determine parameters
Curvature-dependent light speed	Structured hypothesis	Operational formulation and experimental tests
Curvature-driven mass formula	Structured hypothesis	First-principles derivation
Geometric representation of entanglement	Hypothesis	Recover quantum statistics and no-signaling
Proton spin from geometric dynamics	Open proof program	Derive quantization and $J = 1/2$ state selection
Superquantum correlation expression	Conjectural extension	Consistency and experimental discrimination
Sterile neutrino excluded from the proposed WA mode set	Model choice	Do not confuse with a universal experimental exclusion

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The present manuscript does not claim that all equations have been numerically validated or that all proposed predictions have been tested. Code, datasets, and reproducible computational procedures should accompany any future quantitative claim.

2.20 Acknowledgments

We are grateful to all the individual scientists and teams of scientists who have contributed to the exploration and understanding of the entire universe. Supported by AI-driven Supercomputers and the MES Universe Project. We have benefited from our amazing AI partners.

2.21 Funding

Not applicable

2.22 Availability of data and material

Code, mathematical scripts, and preprints are available at GitHub: https://github.com/Zaitian001/-Self-Preprint-V1.1-

2.23 Competing interests

Not applicable

2.24 Author Contributions

Baoliin (Zaitian) Wu 伍宝林 is the sole author of this manuscript and is responsible for all aspects of the research, including conceptualization, methodology, theoretical formulation, and manuscript writing.

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2.26 Physical Proof of Existence (Appendix)

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